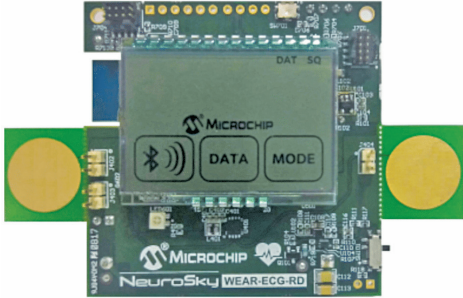


Interesting

Reference Designs OF WEARABLES

Title of Reference Design	Wearable Electrocardiogram	Heart Rate Monitor Wrist Band
Image		
Key Attractions	The reference design showcases advanced ECG data analysis and IoT communication capabilities with devices like smartphones and tablets, featuring multiple operational modes and energy-efficient low-power mode for prolonged use.	The reference design enables a compact, cost-effective, low-power, and accurate heart rate monitor with integrated battery charging, power management, high-resolution signal processing, and connectivity for efficient wearable devices.
Highlights	The wearable ECG reference design from Microchip, equipped with an onboard accelerometer, measures complex heart rate data and other cardiac-related metrics while tracking patient movement. Its ECG demo board assesses the capabilities of the Microchip ATSAML22N18A ARM-based microcontroller and components like the BTLC1000-MR110CA and AT-ECC508A in a wearable ECG context. The board, using Atmel Studio's integrated development environment, offers straightforward access to features and demonstrates integration in an ECG setting. It supports multiple modes of operation, including a default normal mode with sub-modes for ECG, standalone (DEMO), and USB (DATA) for data transfer, and a low-power mode to conserve energy. This mode deactivates unnecessary components and is accessible only from ECG monitoring mode to minimise power consumption. It ensures essential functions like PTC monitoring and RTC with the SleepWalking feature remain active. Additionally, the board's BLE module enables IoT capabilities by facilitating communication with devices like tablets, smartphones, or PCs.	The MAXREFDES1207 from Analog Devices (ADI) is a reference design for wearable applications. This design enables the creation of a compact, cost-effective, low-power, and highly accurate heart rate monitor using two green LEDs. The MAX30112, a specialised pulse oximeter and heart rate analogue front-end (AFE), offers high-resolution optical signal processing with ambient light cancellation and high-current LED driver DACs, making it ideal for wrist-worn devices. The MAX77651B provides integrated battery charging and power management, featuring a SIMO buck-boost regulator for three power rails from a single inductor and a versatile linear charger supporting various Li+ battery sizes. The MAX32660 is an ultra-low-power microcontroller with a flexible power management unit and a powerful ARM Cortex-M4 processor with an FPU, suitable for battery-operated devices. The MAX40005, a compact single comparator, is designed for portable electronics with space and power constraints. The design also includes an FTDI FT234 USB-to-serial UART interface. Key features and benefits include an integrated solution for seamless performance, compact size for portability, low power consumption for extended use, and high accuracy, making it ideal for users seeking efficiency and reliability.
Applications	The reference design is utilised in fitness applications and medical device technology.	The reference design is suitable for wearable health applications.
OEM Brand	Microchip	Analog Devices (ADI)
URL	https://www.electronicsforu.com/electronics-projects/wearable-electrocardiogram-reference-design	https://www.electronicsforu.com/electronics-projects/heart-rate-monitor-wrist-band-reference-design